

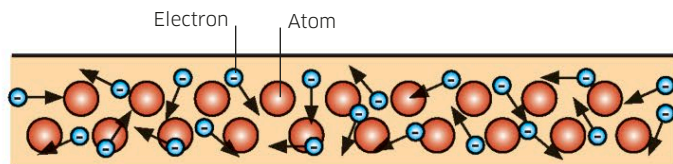
Electricity

A useful form of energy that can be converted to heat, light, and sound, electricity powers the modern world.

Atoms contain tiny particles called electrons that carry negative electrical charge. These orbit the positively charged atomic nucleus, but can become detached. Static electricity is the build-up of charge in an object. Current electricity is when charge flows.

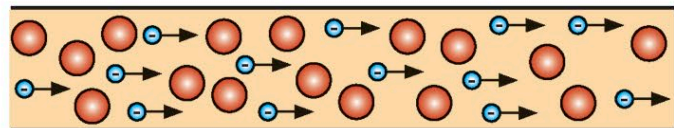
Current electricity

When an electric charge flows through a metal, it is called an electric current. The current is caused by the drift of negatively charged electrons through a conductor in an electrical circuit. Individual electrons actually travel very slowly, but pass electrical energy along a wire very fast.



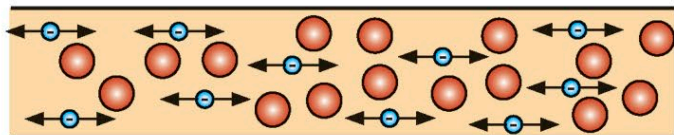
No current

If a conductor wire is not connected to a power supply, the free electrons within it move randomly in all directions.



Direct current (DC)

If the wire is given energy by a battery, electrons drift toward the positive pole of the power supply. If the charge flows in one direction, it is known as direct current (DC).



Alternating current (AC)

Electrical grid electricity runs on an alternating current (AC) supply. The charge changes direction periodically, sending the electrons first one way and then the other.

Copper wire is a good conductor.



Plastic is an insulator.

Conductors and insulators

Charged particles can flow through some substances but not others. In metals, electrons move between atoms. In solutions of salts, ions (positively charged atoms) can also flow. These substances are known as conductors. Current cannot pass through insulators, such as plastic, which have no free electrons. Semi-conductors such as silicon have atomic structures that can be altered to control the flow of electricity. They are widely used in electronics.

Static electricity

Electricity that does not flow is called static electricity. A static charge can be produced by rubbing two materials together, transferring electrons from one to the other. Objects that gain electrons become negatively charged, while objects that lose electrons become positively charged.



Attraction and repulsion

Rubbing balloons against your hair will charge the balloons with electrons, leaving your hair positively charged. The negative charge of the balloons will attract the positive charge of your hair.



Static discharge

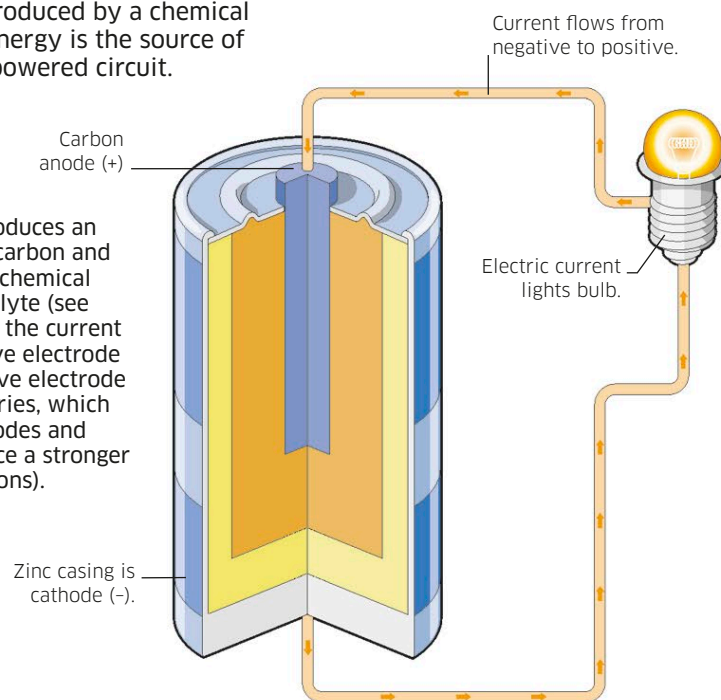
When ice particles within a cloud collide, they gain positive and negative charge. Lightning is an electrical discharge between positive and negative parts of a thunderstorm cloud and the ground.

Making electricity

In order to make electrons move, a source of energy is needed. This energy can be in the form of light, heat, or pressure, or it can be the energy produced by a chemical reaction. Chemical energy is the source of power in a battery-powered circuit.

Battery

A standard battery produces an electric current using carbon and zinc conductors and a chemical paste called an electrolyte (see pp.52-53). In a circuit, the current flows from the negative electrode (cathode) to the positive electrode (anode). Lithium batteries, which have manganese cathodes and lithium anodes, produce a stronger voltage (flow of electrons).



Solar cell

Light falling onto a "photovoltaic" cell, such as a solar cell, can produce an electric current. Light knocks electrons out of their orbits around atoms. The electrons move through the cell as an electric current.